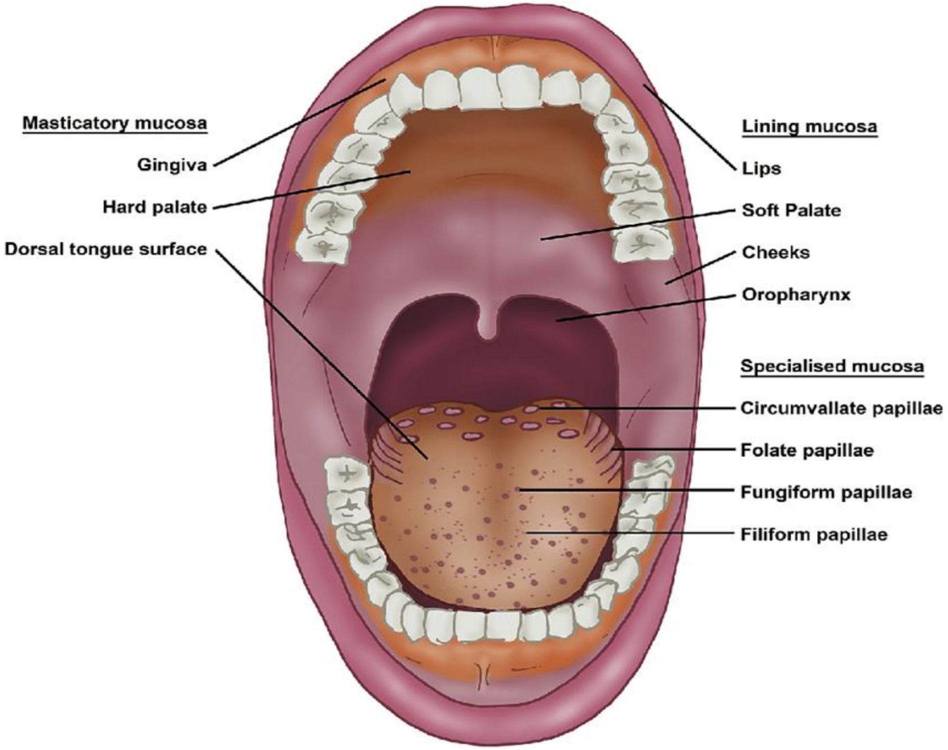
ORAL MUCOUS MEMBRANE



DR SAJDA KHAN GAJDHAR

COURSE CO-ORDINATOR

ORAL CAVITY IN HEALTH

Keratinized Areas

- Masticatory Mucosa
  - Hard Palate
  - Gingiva
- Vermilion Border

Keratinized area

- **Masticatory mucosa:**
  1. Subjected to friction & pressure during mastication.
  2. Firmly attached to bone.
  3. Do not stretch.
- **Vermilion Border:**

  A transient structure between skin and mucosa

GINGIVA

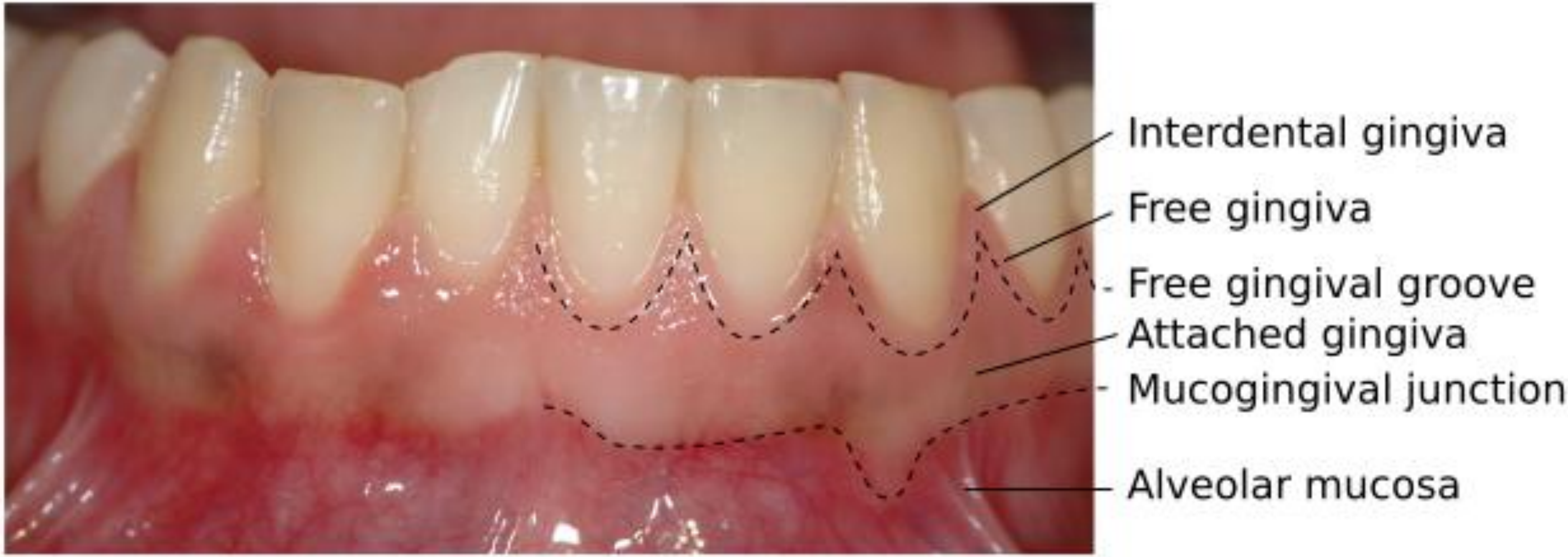

ALVEOLAR MUCOSA
ALVEOLAR MUCOSA
MUCO-GINGIVAL LINE
MUCO-GINGIVAL LINE
ATTACHED GINGIVA
ATTACHED GINGIVA
MARGINAL GINGIVA
MARGINAL GINGIVA

PARTS OF GINGIVA

1.Free or Marginal Gingiva

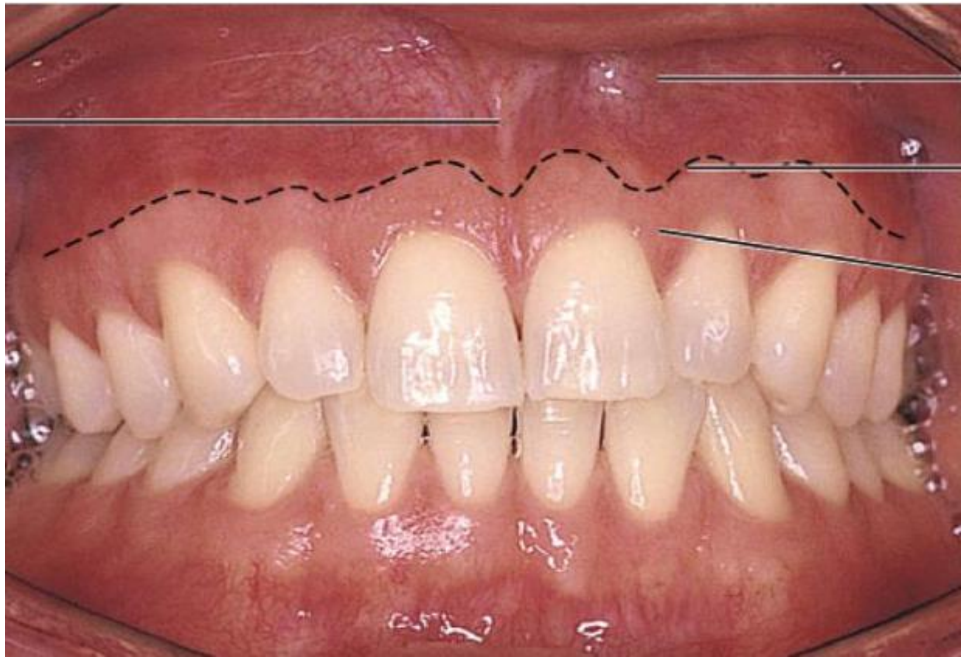
2.Attached Gingiva

3.Interdental Papilla


Interdental gingiva
Free gingiva
Free gingival groove
Attached gingiva
Mucogingival junction
Alveolar mucosa

Free Gingival groove : Demarcation b/w free gingiva and attached gingiva

Mucogingival groove : Demarcation b/w attached gingiva and alveolar gingiva

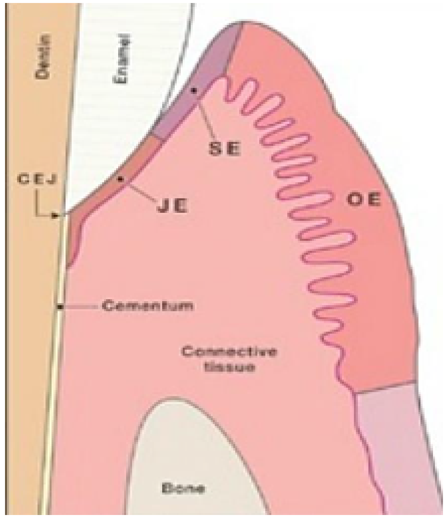


The mucogingival junction disappears with inflammation where the gingiva become red

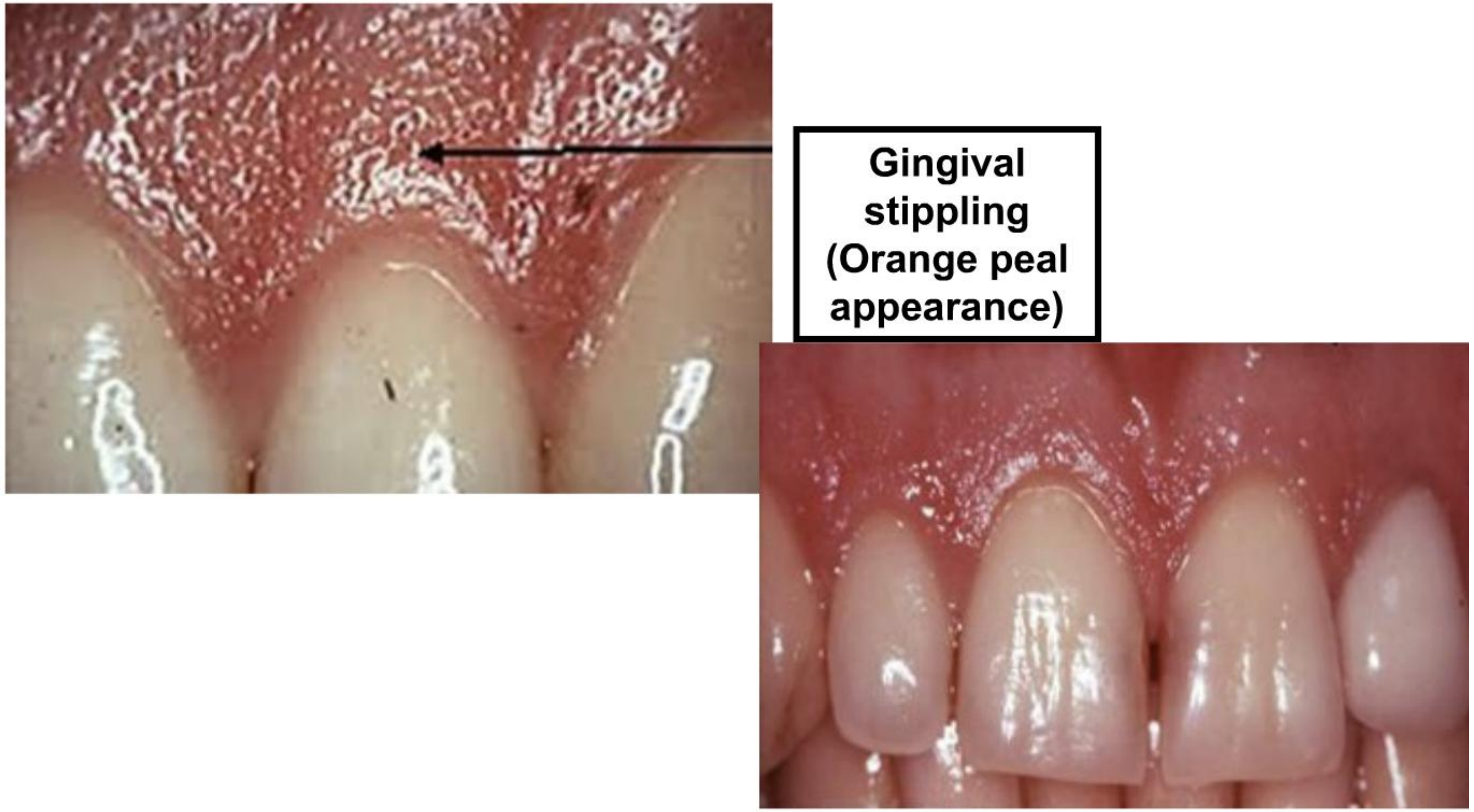
Free Gingiva	Attached Gingiva	Interdental Papilla
Movable part of the gingiva that forms a margin at the neck of the tooth	Attached to the tooth by Junctional epithelium	Part of gingiva present b/w two teeth apical to contact point
Keratinized	Keratinized	
Not Stippled	Stippled	
Bound on the inner margin by gingival sulcus that separates it from the tooth and apically bound by free gingival groove	Present between free gingival groove and alveolar mucosa and separated from alveolar mucosa by Mucogingival junction	Triangular shaped from labial or lingual aspect

Components of Free Gingiva:

- Junctional Epithelium
  - Attached to the tooth by Hemidesmosomes
  - Weak area of Oral mucus membrane
  - Forms floor of gingival sulcus
- Sulcular or Crevicular Epithelium
  - Present b/w Junctional epithelium and Oral epithelium
- Outer Oral Epithelium



Healthy Gingiva

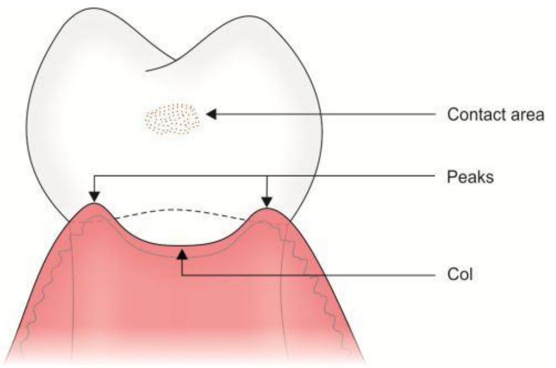


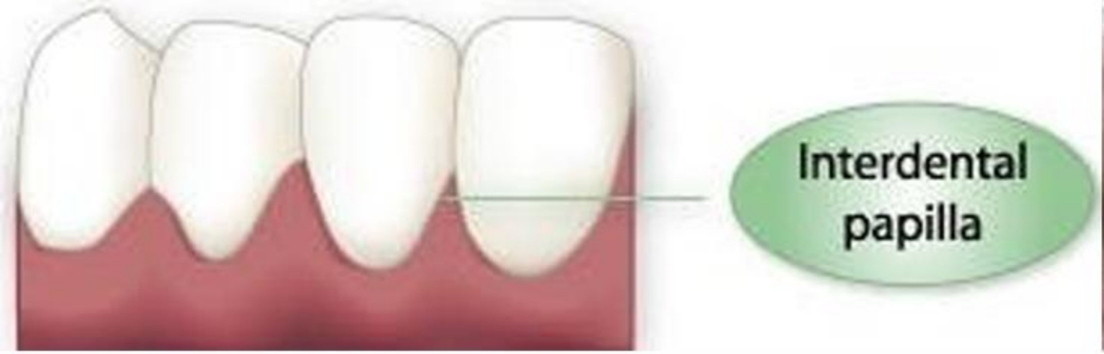
Interdental papilla:

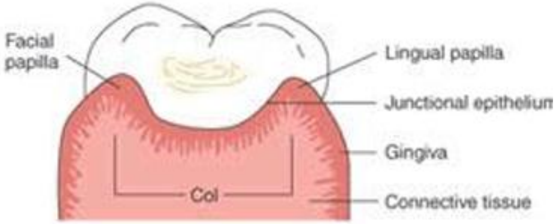
Site :Fill space between 2 adjacent teeth.•

Shape:

1. Triangular from oral or vestibular view.
2. Tent shape (3 dimensional view) with central concave area fits below contact area called interdental col covered by thin non-keratinized epithelium representing the only part of the gingiva covered by Non-keratinized mucosa.



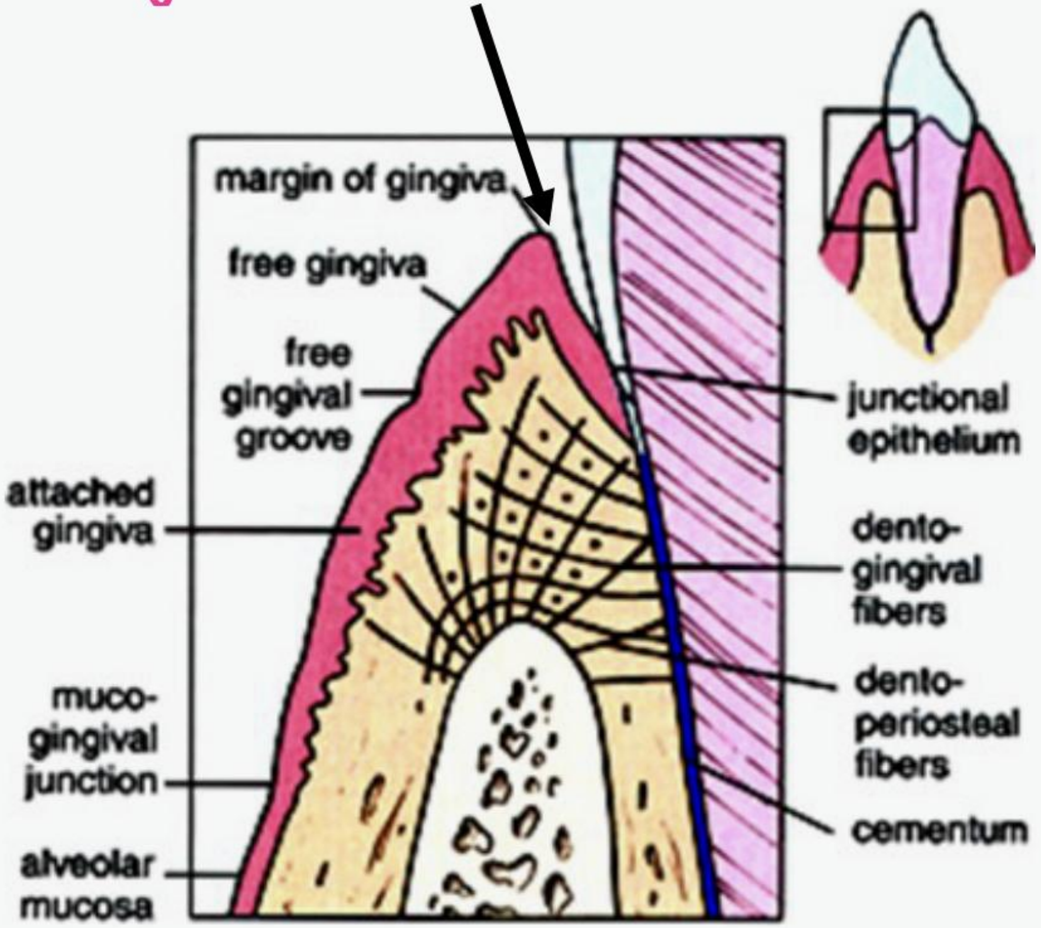


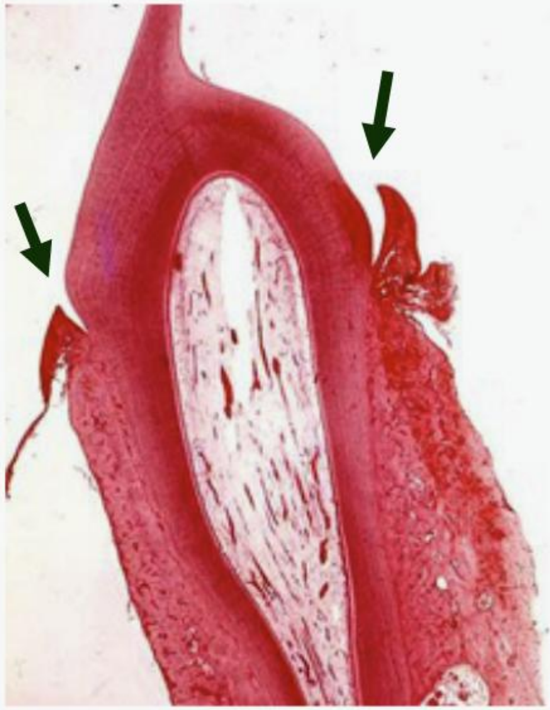


Gingival Sulcus:

Non- Keratinized

Invagination made by the gingiva as it joins tooth surface





- Extension:

From free gingival margin

⇩

Dento-gingival junction

- Average biologic depth

0.5 - 1.8 mm.

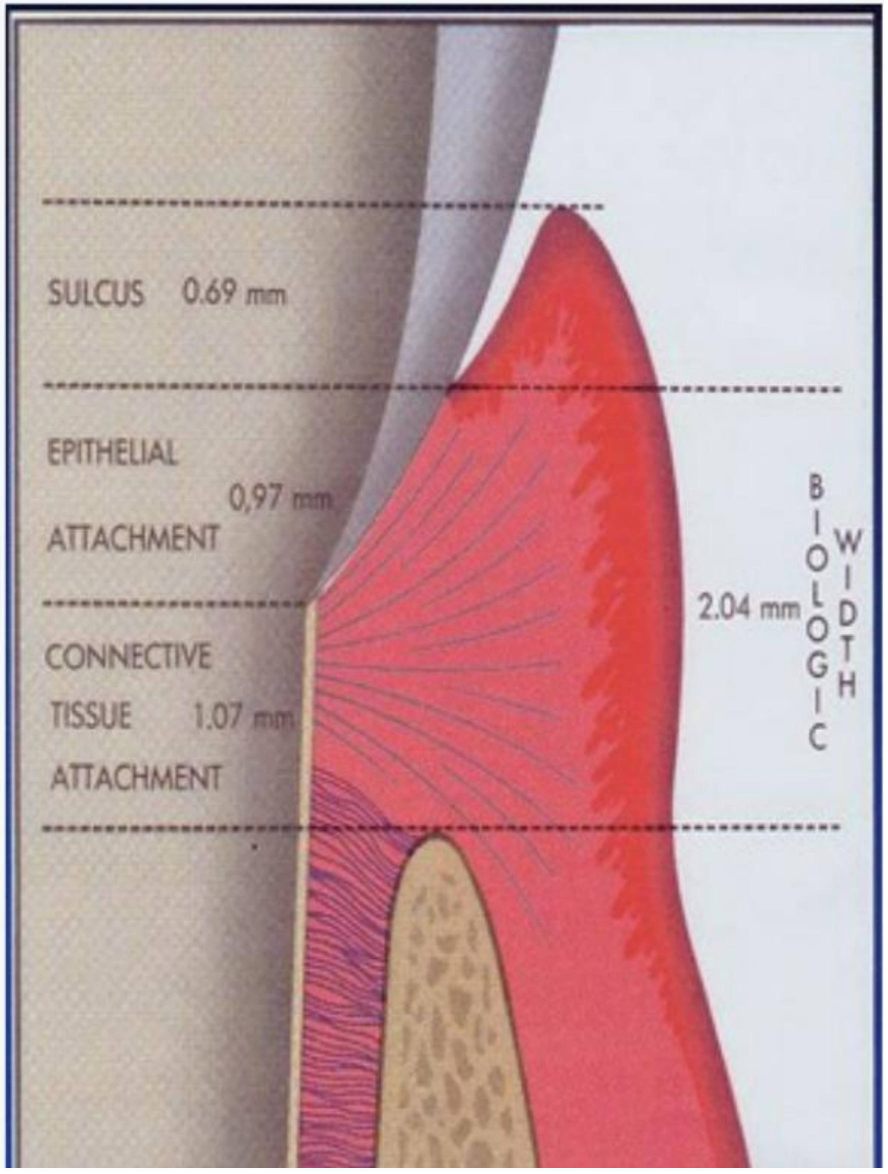
Dento-gingival junction

- DGJ is the junction bt. tooth & gingival tissue (epith. & C.T.)

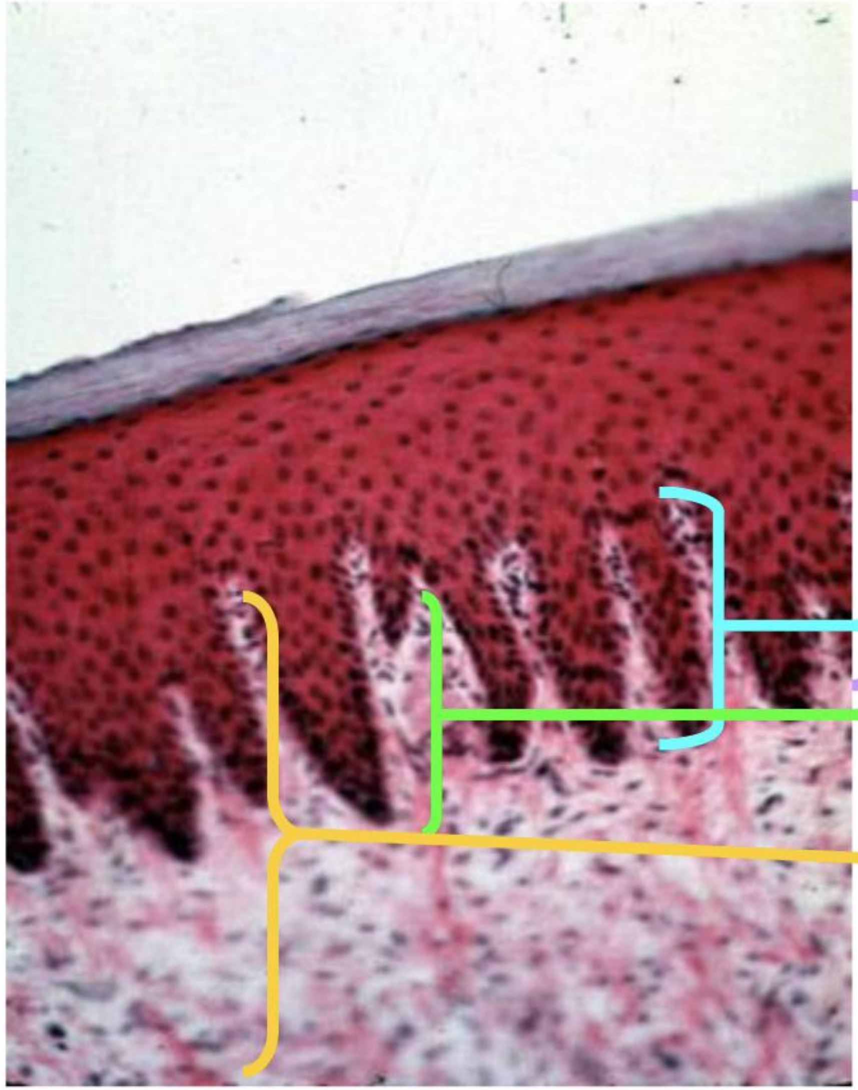
- The epithelial part is called the attachment epith. OR. Junctional epith. Which extend from the base of the sulcus to cemento-enamel junction (forms the floor of the sulcus).

- It is a unique feature of attached gingiva.

- Disturbance of this attachment results in deepening of the sulcus → periodontal problem.



Histology of gingiva



Gingiva is formed of K.St.Sq. epith.
Except 2 regions:
1-Gingival sulcus
2-Interdental Col
non-keratinized

Stratified squamous keratenized epithelium

Epithelial rete peg

C.T papilla

Tall

Numerous

Slender

Irregular

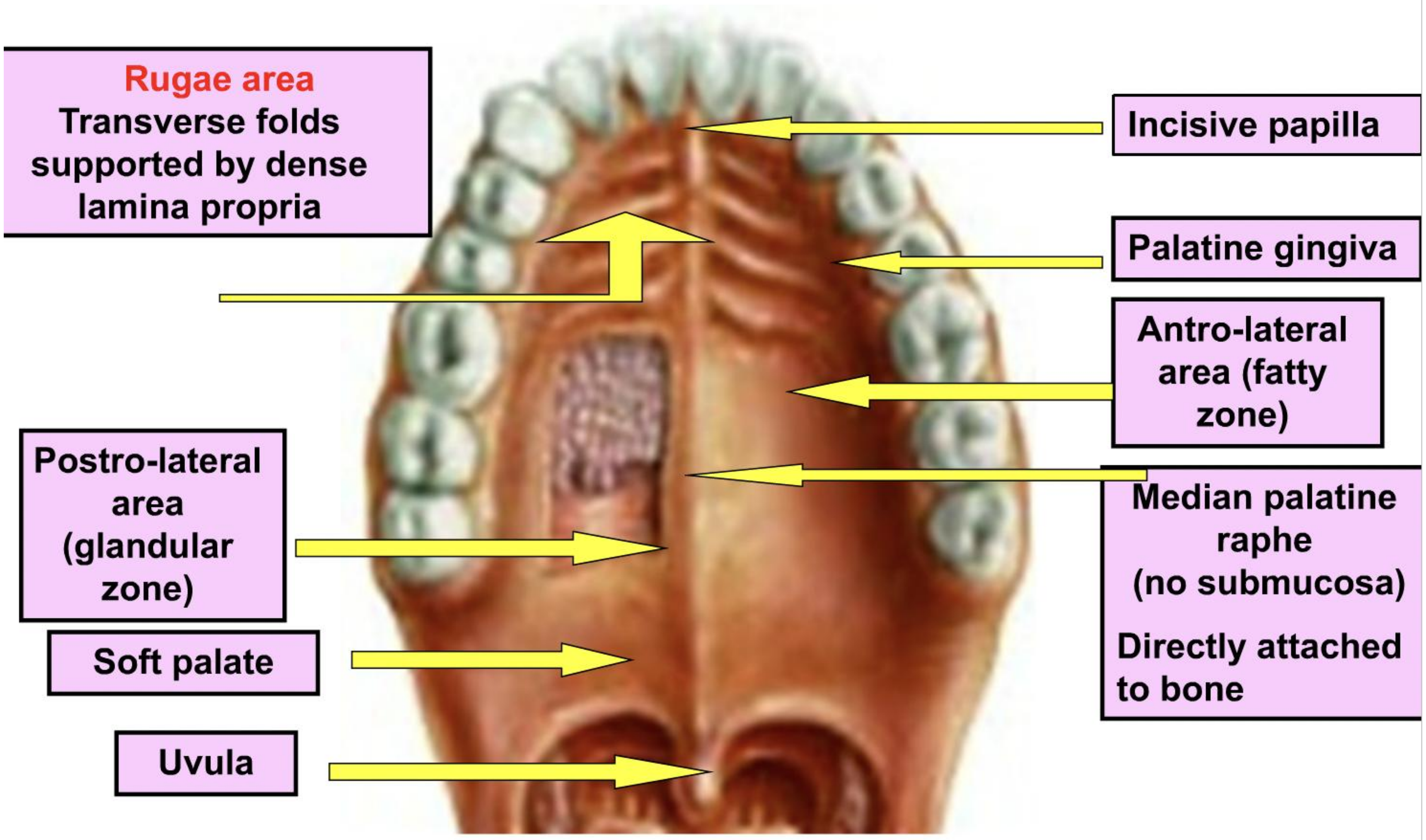
Lamina propria

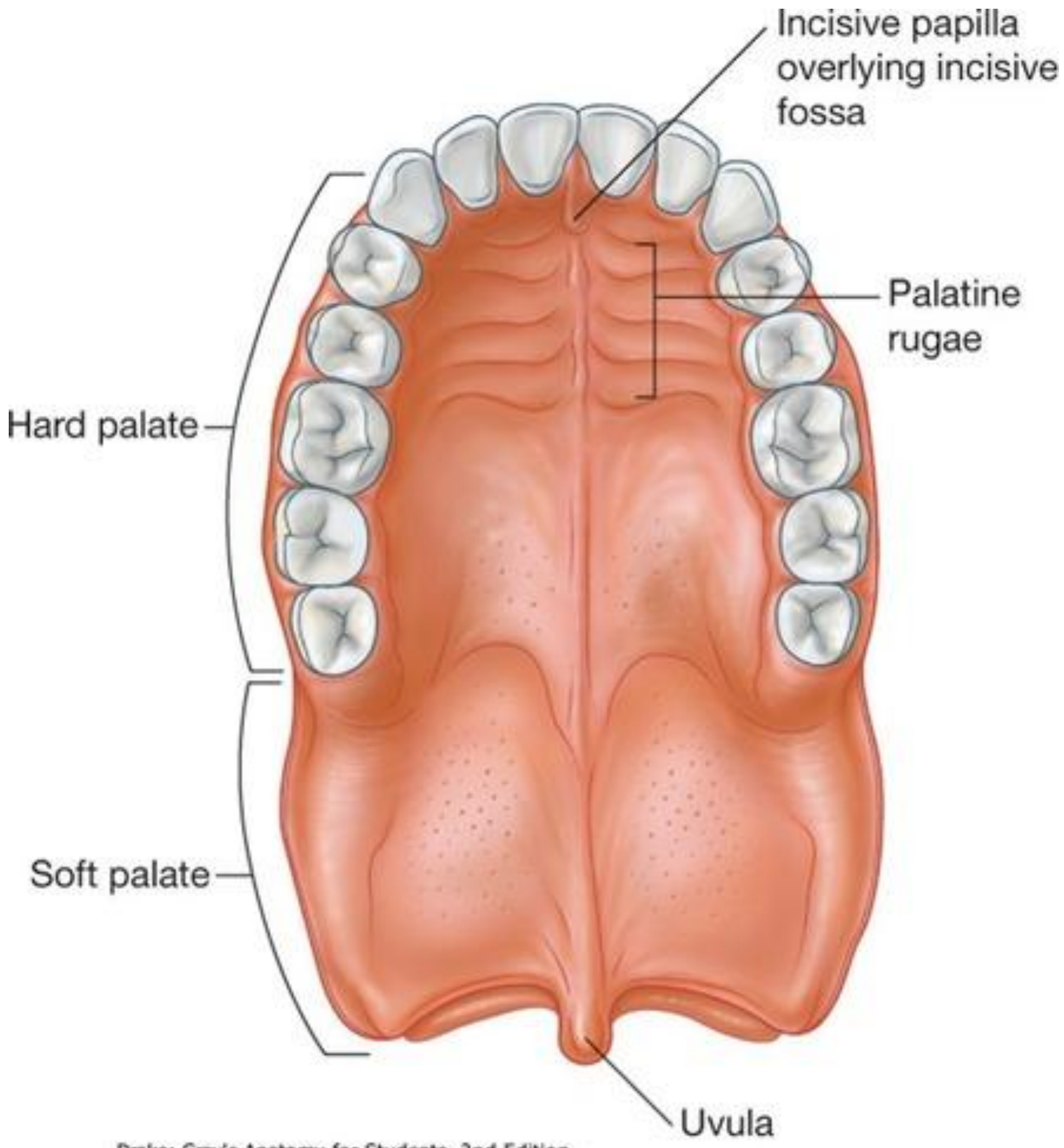
No submucosa

- **Epithelium**
- Outer Oral epithelium: Keratinized stratified squamous epithelium (75% Parakeratinized) with long and sharp rete ridges
- Sulcular epithelium: Nonkeratinized
- Junctional epithelium: Nonkeratinized
- Col: Nonkeratinized

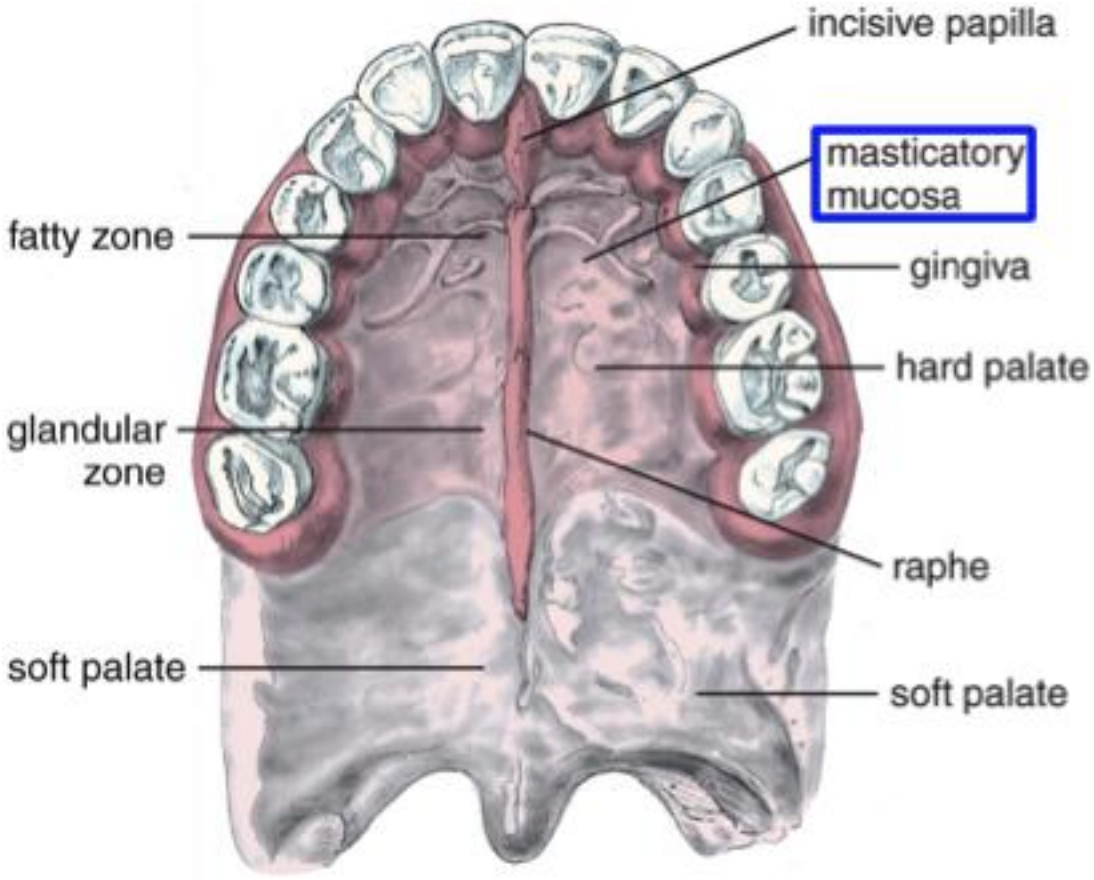
- **Lamina Propria :** dense bundles of collagen fibres
- **Submucosa :** absent

Macroanatomy of palate



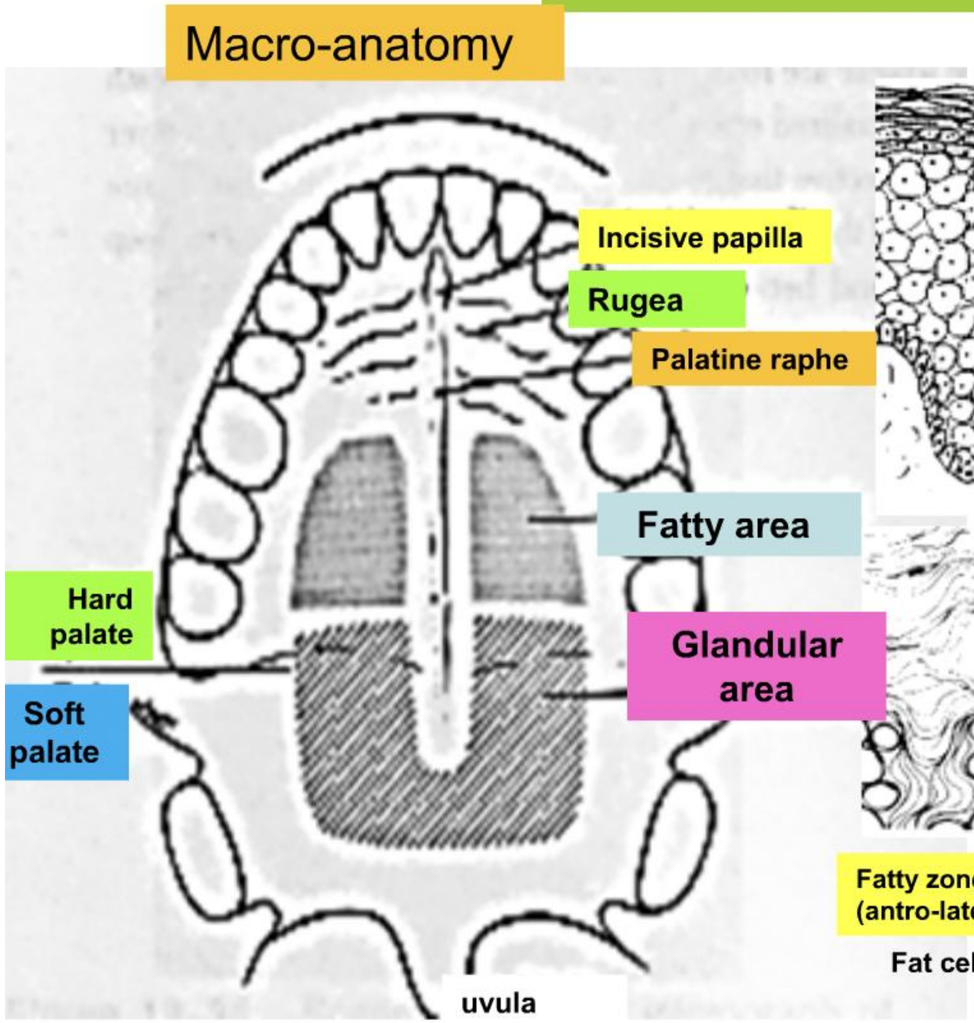


Drake: Gray's Anatomy for Students, 2nd Edition.

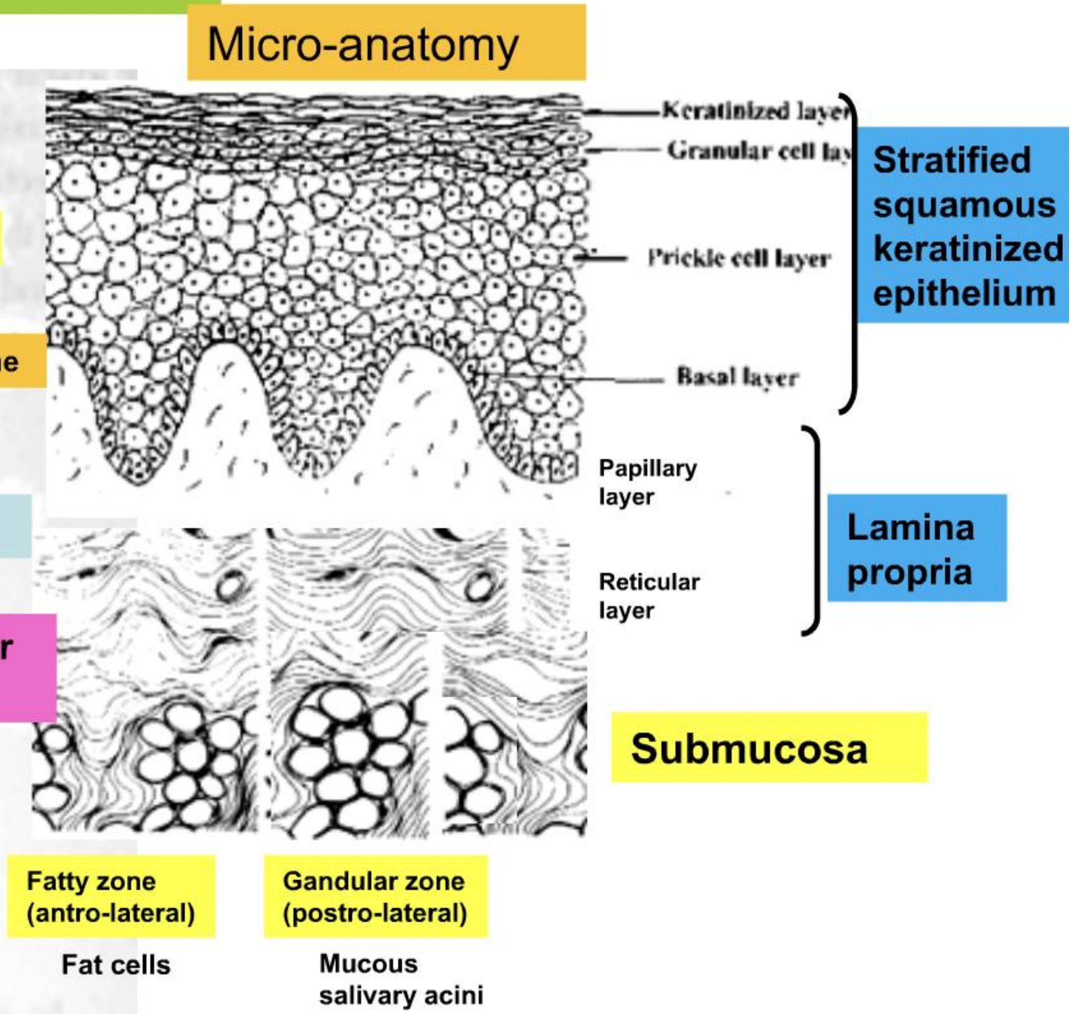


Hard Palate Macro-anatomy

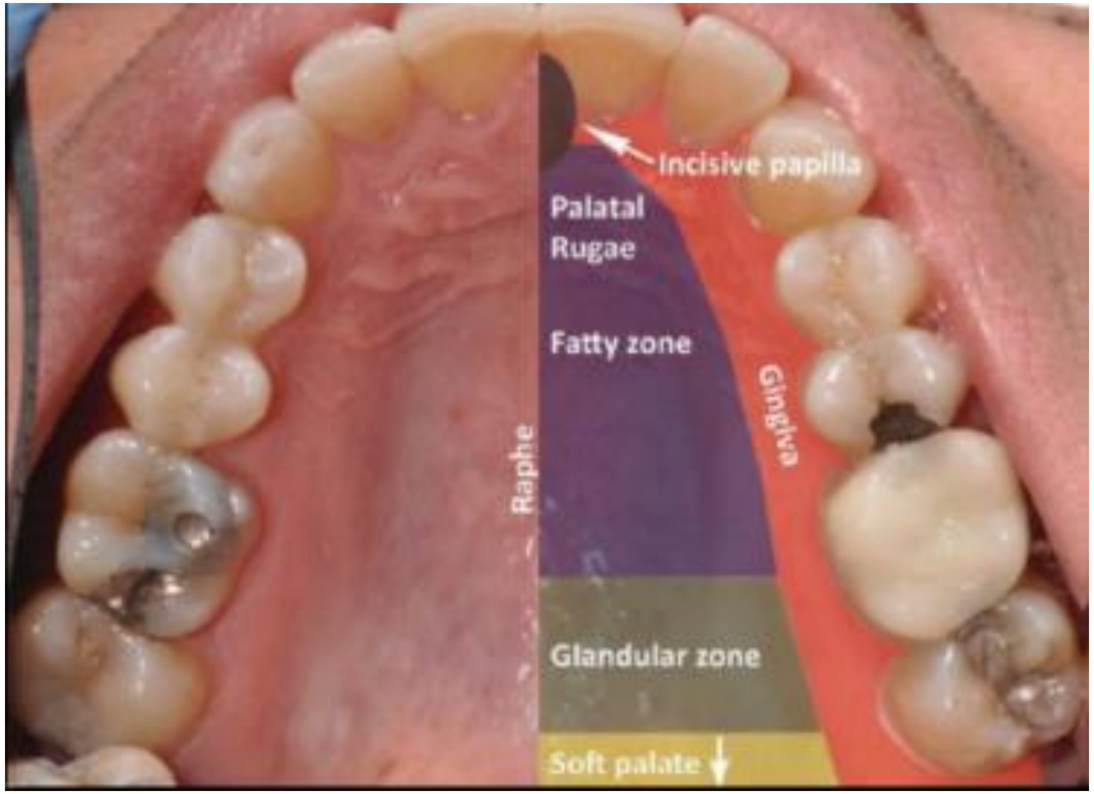
Hard Palate



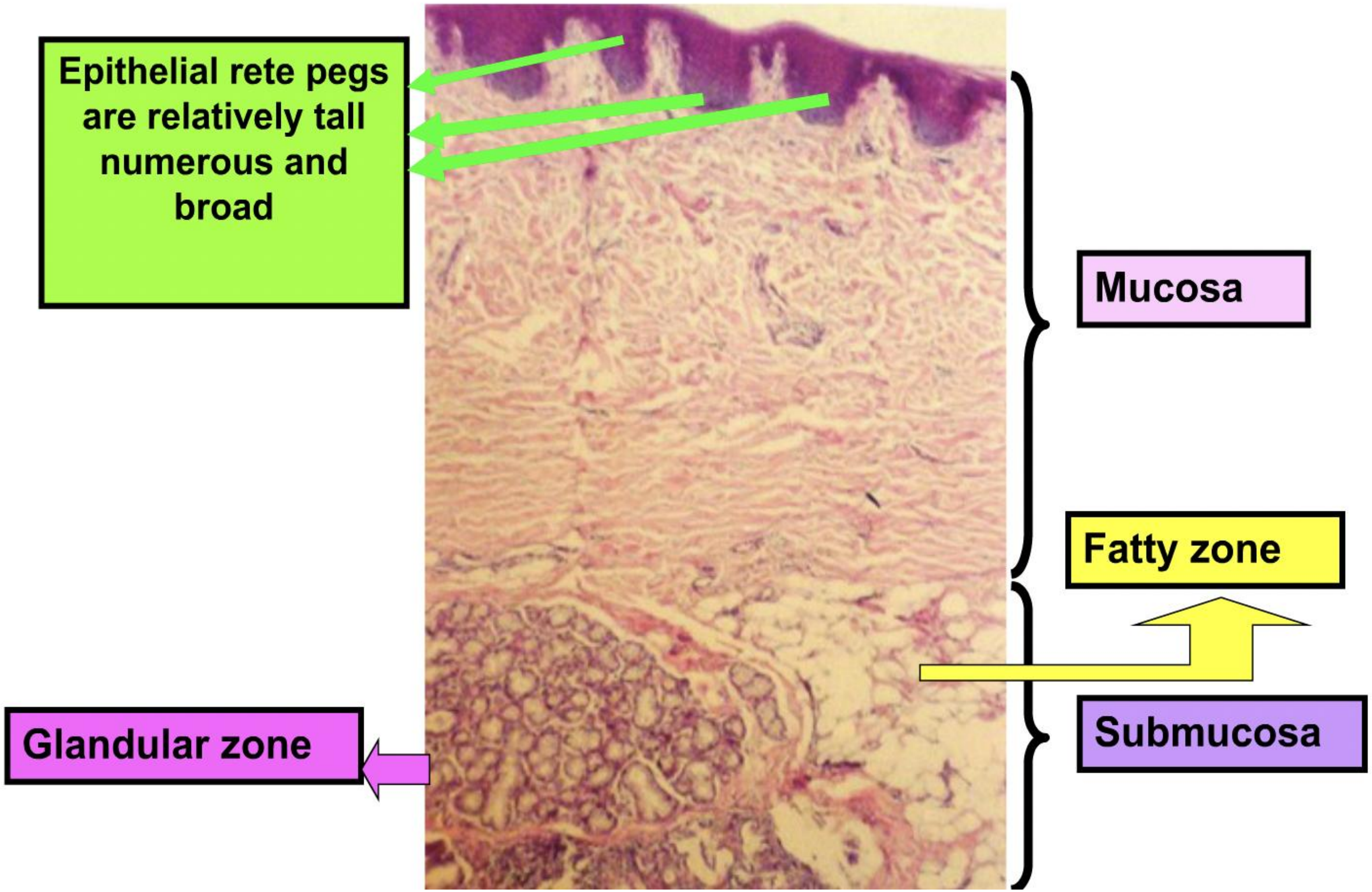
Macro-anatomy
Incisive papilla
Rugea
Palatine raphe
Fatty area
Hard palate
Glandular area
Soft palate
uvula



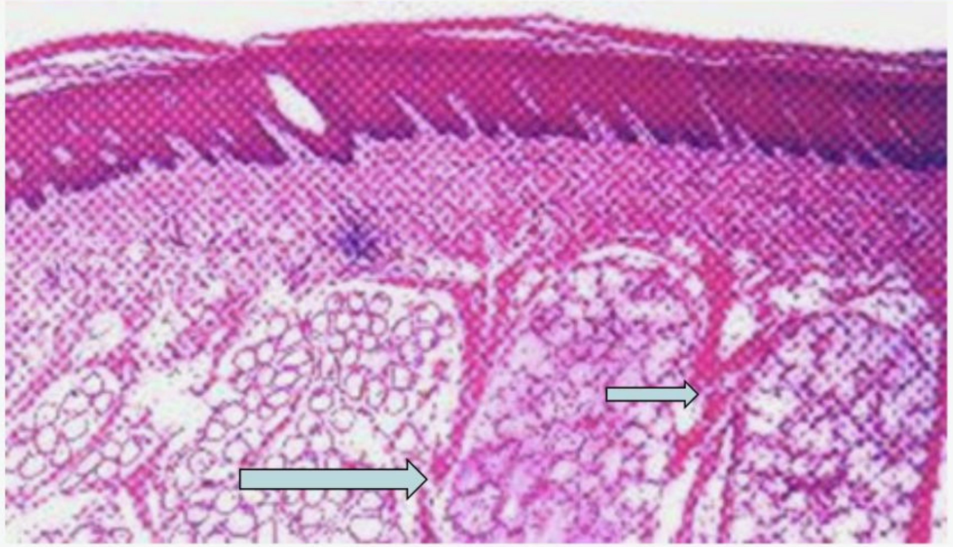
Micro-anatomy
Keratinized layer
Granular cell lay
Prickle cell layer
Basal layer
Stratified squamous keratinized epithelium
Papillary layer
Reticular layer
Lamina propria
Submucosa
Fatty zone (antro-lateral)
Fat cells
Gandular zone (postro-lateral)
Mucous salivary acini


Incisive papilla
Palatal
Rugae
Fatty zone
Gingiva
Raphe
Glandular zone
Soft palate

Histology of hard palate



Hard Palate Micro-anatomy



Traction bands are bundles of collagen fs. that extend from of the lamina propria to the underlying bone, dividing the submucosa into compartements(anterolateral part contain fat cells & postrolateral contain mucous salivary glands)

- Oral Development and Histology Avery, j. K.1st edition.
- Orban's Oral Histology and Embryology Bhaskar, s. N.11th edition.
- Oral Histology : Development, Structure and Funct Tencate, a. R. 4th edition.